Muhammad Muizz Zafar

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Skills

Programming Languages:	Python, SQL, C#, Java, JavaScript, C++, HTML/CSS
Databases:	MS SQL Server, Oracle, PostgreSQL, Relational Databases
Tools/Frameworks:	PySpark, ETL Pipelines, Power BI, Power Automate, AWS, Docker, Flask, React

Education

University of Ottawa — Ottawa, ON
Bachelor of Applied Science, Software Engineering *GPA: 3.7/4.0* — Sept 2020 - Dec 2024

Work experience

Software & Database Analyst — May 2024 - Present
Elections Canada — Ottawa, ON

- Executed enterprise database migration upgrading **SQL Server 2019 to SQL Server 2022** across **DEVTEST, UAT, and PROD** environments, migrating **50M+ records**, schemas, stored procedures, triggers, and security configurations with zero downtime.
- Implemented a reusable ETL validation framework using **Python** and **PySpark** to validate and monitor **data pipelines**, improving data quality across multiple environments.
- Leveraged **Power Automate** to orchestrate ETL workflows by triggering **Python**-based validation scripts, automating testing cycles, and sending pass/fail notifications with linked reports to stakeholders.
- Developed **Power BI** dashboards to visualize database health, operational metrics, and data quality trends, enabling data-driven decision making for senior management.
- Performed **SQL-based data analysis and validation** across relational databases (**MS SQL Server** and **Oracle**), identifying discrepancies and supporting ETL issue resolution.
- Coordinated version-controlled **database deployments** and release processes, ensuring traceability and reliability across shared environments.
- Collaborated within an **Agile cross-functional team** to troubleshoot data issues, enhance ETL workflows, and improve system reliability.

Software Developer Intern — September 2023 - April 2024
University of Ottawa — Ottawa, ON

- Supported development and maintenance of **Microsoft Power Platform applications**, including **Power Automate**, to enhance workflow automation, data processing, and operational efficiency.
- Developed **Power Platform solutions** including **Canvas Apps and Model-Driven Apps** integrated with **C# backend services** to support event management, user interactions, and data-driven workflows
- Implemented automated workflows using **Power Automate**, leveraging **HTTP triggers, scheduled flows, and API integrations** to enable real-time notifications, approvals, and data synchronization across systems
- Designed end-to-end integrations between **Power Apps and REST APIs**, enabling seamless communication between frontend applications and backend services
- Redesigned the University of Ottawa's emergency alert website with **C# (.Net Core)** and **Angular**, introducing a modern, interactive UI. Used by over **20,000** students daily. website: ue.uottawa.ca/
- Optimized existing **C#** REST APIs by reducing **LINQ** query execution time for event data lookup by **30 %** and lines of code by **40 %**
- Participated in end-to-end **Software Development Lifecycle (SDLC)** processes, from requirements gathering and system design to implementation, testing, and production deployment

Cloud Developer Intern — May 2023 - August 2023
Trend Micro Inc. — Ottawa, ON

- Utilized **AWS Lambda** to monitor client files, images and code uploaded to the cloud and scan for vulnerabilities using Anchore APIs and Grype
- Programmed robust error handling for **AWS Lambda** functions in GoLang using goroutines to ensure proper timeout and recovery, reducing data corruption and improving service reliability

Cloud Developer Intern — September 2022 - December 2022
Trend Micro Inc. — Ottawa, ON

- Developed a serverless automation solution using **AWS Lambda, S3, and Athena crawlers** to trigger metadata updates on data changes, significantly reducing manual intervention and operational overhead
- Configured **Splunk** and **AWS CloudWatch** alarms for proactive monitoring of test logs and crashes, enhancing system reliability and reporting.

Full Stack Developer Intern — January 2022 - April 2022
Applied Mind Inc. — Ottawa, ON

- Utilized containerization concepts to create **Docker** containers for the full- stack application for PROD deployment
- Automated the event detection process by adding events to **PostgreSQL** DB using **Python** and **GraphQl** which helped reduce testing time by **10 seconds**